

Muhsin Aljaf

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Employment History

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| June. 2023 – Present | ◇ Research Assistant , Department of Physics, Oakland University, Michigan, USA |
| Jan. 2023 – May | ◇ Teaching Assistant , Department of Physics, Oakland University, Michigan, USA |
| Nov. 2020 – Dec 2022 | ◇ Assistant lecturer , Department of Physics, School of Science, University of Garmian, University of Garmian, Kurdistan region, Iraq. |
| June. 2020 – Nov. 2020 | ◇ Lab Assistant , Department of Physics, School of Science, University of Sulaimani, Kurdistan Region, Iraq. |
| Sept. 2015 – Aug. 2017 | ◇ Lab Assistant , Department of Physics, School of Science, University of Sulaimani, Kurdistan Region, Iraq. |

Education

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| Jan 1, 2023 – Present | ◇ Ph.D. in Applied and Computational Physics , Department of Physics, Oakland University, Rochester, MI 48309, USA.
<i>Relevant Coursework:</i> <ul style="list-style-type: none">• PHY 5620 - Mechanics II• PHY 6740 - Advanced Quantum Mechanics• PHY 5830 - Classical Electrodynamics• PHY 5900 - ST: Cosmology• PHY 5900 - ST: General Relativity• SCI 5110 - Ethics and Practice of Science• PHY 5520 - Theoretical Physics• PHY 6730 - Quantum Mechanics• STA 5001 - Statistical Methods in Research and Production |
| 2018 – 2020 | ◇ M.Sc in Astrophysics , University of Science and Technology of China.
Thesis title: <i>Phase space analysis and singularity classification for linearly interacting dark energy models.</i>
◇ School On CMB Physics and QFT , University of Science and Technology of China, Hefei, China, Nov 2018. |

Education (continued)

- ◇ **Summer School on Particle Physics**, International Centre for Theoretical Physics (ICTP), Trieste, Italy, Jun 2019.
- 2017 – 2018 ◇ **1 Year Chinese language course**, Anhui Normal University, Wuhu, China.
- 2010 – 2014 ◇ **B.Sc. in Physics**, University of Sulaimani, Sulaimaniyah, Kurdistan Region, Iraq.
Thesis title: *The origin and distribution of life in the universe.*
- 2004 – 2010 ◇ **Garmian High School**, Major in Science
- 1998 – 2004 ◇ **Kawacharmu Elementary School**

Research Publications

- 1 Aljaf, M., & Cholis, I. (2025a). The Merger Rate of Primordial Black Holes. arXiv: 2512.12227 [astro-ph.HE]
- 2 Bouhaddouti, M. E., Aljaf, M., & Cholis, I. (2025). Conservative limits on primordial black holes from the LIGO-Virgo-KAGRA observations. arXiv: 2502.00144 [astro-ph.CO]
- 3 Aljaf, M., & Cholis, I. (2025b). Simulating binary primordial black hole mergers in dark matter halos. *Phys. Rev. D*, 111(6), 063020. doi:10.1103/PhysRevD.111.063020. arXiv: 2408.06515 [astro-ph.GA]
- 4 Aljaf, M., Elizalde, E., Khurshudyan, M., Myrzakulov, K., & Zhadyranova, A. (2022). Solving the H_0 tension in $f(T)$ gravity through Bayesian machine learning. *Eur. Phys. J. C*, 82(12), 1130. doi:10.1140/epjc/s10052-022-11109-y. arXiv: 2205.06252 [astro-ph.CO]
- 5 Aljaf, M., Gregoris, D., & Khurshudyan, M. (2022). Assessing the foundation and applicability of some dark energy fluid models in the Dirac–Born–Infeld framework. *Int. J. Mod. Phys. A*, 37(34), 2250211. doi:10.1142/S0217751X22502116. arXiv: 2010.05278 [gr-qc]
- 6 Aljaf, M., Gregoris, D., & Khurshudyan, M. (2021). Constraints on interacting dark energy models through cosmic chronometers and Gaussian process. *Eur. Phys. J. C*, 81(6), 544. doi:10.1140/epjc/s10052-021-09306-2. arXiv: 2005.01891 [astro-ph.CO]
- 7 Aljaf, M., Gregoris, D., & Khurshudyan, M. (2020). Phase space analysis and singularity classification for linearly interacting dark energy models. *Eur. Phys. J. C*, 80(2), 112. doi:10.1140/epjc/s10052-020-7671-x. arXiv: 1911.00747 [gr-qc]
- 8 Qiu, T., Xiao, Z., Shi, J., & Aljaf, M. (2020). Potential-driven Inflation with Disformal Coupling to Gravity. *Phys. Rev. D*, 102(6), 063506. doi:10.1103/PhysRevD.102.063506. arXiv: 2004.14712 [hep-ph]
- 9 Hossieni, H., Jabar, M., Burhan, M., Abdulqadr, I., & Jaza, Z. (2016). A study of fluctuations in sun's radiation by measuring earth's angular velocity. *J Climatol Weather Forecasting*, 4(151), 2.

Talks

- ◇ *The Merger Rate of Primordial Black Holes*, Pizza Talk, Oakland University, Nov 14, 2025.
- ◇ *Primordial Black Hole Merger Rates from LIGO-Virgo-KAGRA Observations*, 11th Gulf Coast Gravity Meeting, University of Mississippi, Apr 11, 2025.

Talks (continued)

- ◇ *Primordial Black Holes in Dark Matter Halos: Merger Rate Simulations and Limits from the LIGO-Virgo-KAGRA Observations*, APS Global Physics Summit, Anaheim, Mar 20, 2025.
- ◇ *On the Merger Rates of Primordial Black Holes in Dark Matter Halos*, Garmian University, Feb 24, 2025.
- ◇ *Primordial Black Holes in Dark Matter Halos: Merger Rate Simulations and Limits from the LIGO-Virgo-KAGRA Observations*, 34th Midwest Relativity Meeting, University of Michigan, November 16, 2024.
- ◇ *Simulating the Primordial Black Hole Binaries Formed in Dark Matter Halos*, 2024 APS April Meeting, Sacramento, April 6, 2024.
- ◇ *"Achievements in Cosmology: An Introduction to Standard Model of Cosmology"*, 2nd ICTP PWF Summer School on Particle Physics and Cosmology, University of Salahaddin, Erbil, Sep. 2021.
- ◇ *"Achievements in Cosmology: An Introduction to Standard Model of Cosmology"*, 2nd ICTP PWF Summer School on Particle Physics and Cosmology, University of Salahaddin, Erbil, Sep. 2021.
- ◇ *"Cosmology with Strong Gravitational Lensing"*, International workshop on the Early Universe, University of Garmian, Kalar, Iraq, Mar. 2021).
- ◇ *"No evidence of Beyond-Soft-Corrections to the Born-Infeld Paradigm in the Cosmic Evolution"*, 2021 Annual Particle Cosmology Workshop, Department of Astronomy, University of Science and Technology of China, Jan. 2021.
- ◇ *"Open Issues in Cosmology"*, 1st ICTP PWF Summer School on Particle Physics and Cosmology, University of Sulaimani, Sulaimaniyah, Dec. 2019.
- ◇ *"Higgs Inflation with a Running Kinetic Term"*, International Center for Relativistic Astrophysics Network, Pescara, Italy, July 2019.
- ◇ *"Model-Independent of Dark Energy Models Using Gaussian Process"*, Particle Cosmology Group, University of Science and Technology of China, Hefei, China, Dec. 2018.
- ◇ *"The Strangeness of Quantum Mechanics and Common Sense"*, Tshik University, Erbil, Iraq, Nov. 2016.
- ◇ *"Origin and Distribution of Life in the Universe"*, Department of physics, University of Suliamani, Sulaimaniyah, Iraq, May 2014.
- ◇ *"The Basic Principles and Applications of Holography"*, Department of Physics, University of Sulaimani, Sulaimaniyah, Iraq, Apr. 2013.

Skills

Languages	◇ Kurdish: Native English: Advanced (TOEFL: 98) Persian: Advanced Arabic: Basic Mandarin Chinese: Intermediate (HSK4)
Coding	◇ Python, Fortran 90, $\text{\LaTeX}$
Computing Software	◇ Wolfram Mathematica, Maple
Web Dev	◇ Wordpress
Operating Systems	◇ Linux, Windows
Packages	◇ Emcee, CSOMOMC, xAct
General	◇ LibreOffice, MS office, Adobe Photoshop, Adobe Premiere

Conferences

- ◇ The Fourth Zeldovich Virtual Meeting, ICRAnet, Pescara, Italy, Sept. 2020.
- ◇ Spring Workshop on Gravity and Cosmology, Kraków, Poland, Jagiellonian university, May 2020.
- ◇ Symposium on "Gravity and cosmology in the Gravitational-Wave Era", Central China Normal University, Wuhan, China (Participated with A Poster on "Non-minimal Higgs inflation"), April 2019.
- ◇ S. -T. Yau Science Forum, Mathematical Science Center, Tsinghua University, Nov 2018.

Miscellaneous Experience

Awards and Scholarships

- 2020 ◇ The ANSO Scholarship For Young Talents, University of Science and Technology of China (I declined this offer because I could not go to the university due to the travel ban of international students imposed by China).
- 2019 ◇ ICTP's Travel Grant for 2019 Summer School Program on Particle Physics.
- 2017-2020 ◇ Chinese Government full Scholarship for my master studies, University of Science and Technology of China.
- 2014 ◇ Department Prize for Outstanding Student Performance, University of Sulaimani.

Outreach

- 2014-Present ◇ **Physickurd**
I value and enjoy giving public talks and writing articles. I am a writer at Physickurd, an online physics outreach blog written by me and some of my Kurdish colleagues. We aim to enhance the general public's awareness of current scientific achievements. Click here to see our content.
- 2019-Present ◇ **Physics Without Frontiers**
I am currently the regional coordinator for Physics Without Frontiers (PWF) at the International Center for Theoretical Physics (ICTP). PWF works to motivate, train, and educate physics and mathematics university students worldwide. Our groups aim to improve the scientific capacity and opportunity for young people, particularly those from groups poorly represented in the scientific field. I have co-organized annual summer schools on cosmology and particle physics in the Kurdistan region. Click here to see more details of our activities.
- 2016-Present ◇ **Chawg Podcast**
I am often invited to talk in Chawg Podcast about Astronomy and cosmology; Chawg is a non-profitable podcast in Kurdish broadcasting from Copenhagen, Denmark aiming to raise the public's awareness of science and technology. Below is a list of my talks.
 - *Special episode: Existence and nothingness*, Chawg Podcast, Jul 2020.
 - *Scientific awareness and dissemination to raise quality of thinking*, Chawg Podcast, May 2017.
 - *The Big bang Theory*, Chawg podcast, Jul 2017.
 - *How is the age of the universe, stars and planets estimated?*, Chawg podcast, Aug 2019.

Teaching Experience

- ◇ **Assistant lecturer**, Department of Physics, University of Garmian, Spring 2021.
Electricity and Magnetism course for 45 physics undergraduate students; wrote and graded exams.
- ◇ **Lab assistant**, Department of Physics, University of Garmian, Spring 2021.
Practical classical Mechanics Laboratory for undergraduate students; assisted students in setting up experiments; graded and evaluated their weekly reports.
- ◇ **Teaching Assistant**, Department of Physics, University of Sulaiamani, Fall 2015.
Practical Java programming for undergraduate students; resolved and fixed the confusion of the students in the lab with their code; wrote and graded exams.

Scientific Administrative Experience

Scientific Organization

- 2022 ◇ Coordinating PWF Kurdistan Region of Iraq: Workshop on Large Hadron Collider, University of Garmian, Kurdistan Region, Iraq.
- 2021 ◇ Coordinating second PWF-ICTP Summer School on Particle Physics and Cosmology, University of Salahaddin-Erbil, Kurdistan Region, Iraq.
- 2019 ◇ Coordinating first PWF-ICTP school on Particle physics and Cosmology, University of Sulaimani, Kurdistan Region, Iraq.

References

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